Problem 13 | Problems T4_PhaseDiagrams I.pdf #13 | Ti-Si · congruent melting

Compounds and Cooling of a Ti-30 % Si Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 5

13. Given the following Ti-Si phase diagram answer the next questions: a) Indicate the phases or compounds with incongruent and congruent melting. b) Write down the invariant reactions, specifying their name, temperature, and composition of the phases present. c) For an alloy with 30% Si describe the solidification and cooling sequence to 600 °C. Describe specifically the microstructure that can be expected at 1329 °C, 1169 °C, and 600 °C. d) For the same alloy of part c) and the temperatures of 1329 °C and 1169 °C indicate phase composition, phase fractions and quantity of microconstituents present.

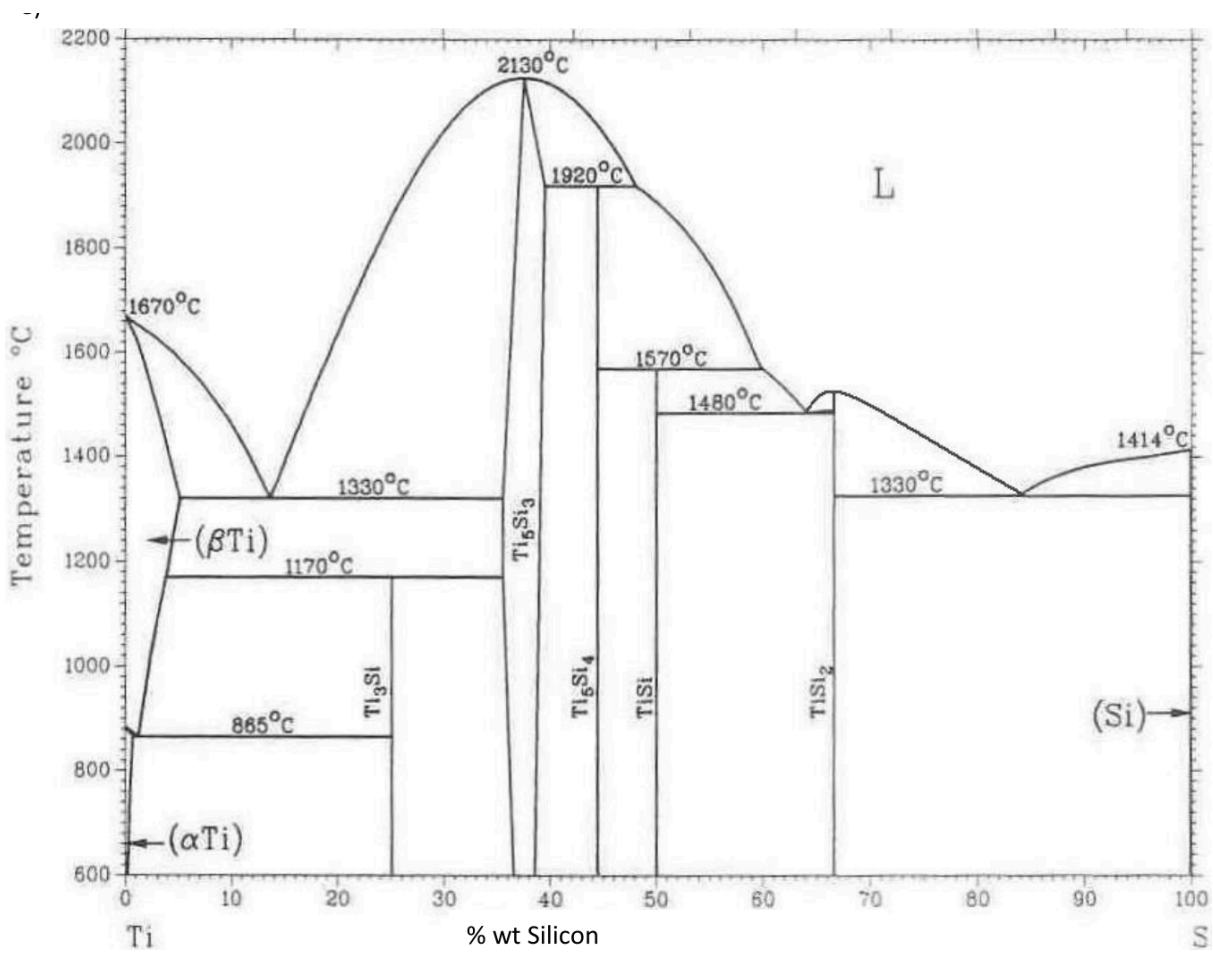


Figure 22. Ti-Si phase diagram (Problems T4_PhaseDiagrams I.pdf, p. 6).

Official Solution: a) congruent melting: Ti_5Si_3 and $TiSi_2$; incongruent melting: Ti_3Si, Ti_5Si_4 and TiSi. b) Eutectic, T = 1330 °C; Eutectic, T = 1330 °C; Eutectic, T = 1480 °C; Peritectic, T = 1920 °C; Peritectic, T = 1570 °C; Peritectoid T = 1170 °C; Eutectoid, T = 865 °C. c) T = 1329 °C: $C_{\beta Ti}$ = 5.5% Si and $C_{Ti_5Si_3}$ = 35% Si, $x_{\beta Ti}$ = 0.17, $x_{Ti_5Si_3}$ = 0.83; T = 1169 °C: C_{Ti_3Si} = 25% Si and $C_{Ti_5Si_3}$ = 35% Si, $x_{Ti_5Si_3}$ = 0.5, x_{Ti_3Si} = 0.5. d) T = 1329 °C: $x_{eutectic}$ = 0.24, $x_{Ti_5Si_3,primary}$ = 0.76; T = 1169 °C phases and microconstituents are the same

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**, Ti-Si diagram of the sheet. Printed temperatures: 1670, 1414, 2130, 1920, 1570, 1480, 1330 (twice), 1170 and 865 °C. Compositions are **readings of the figure** (±1 %), on the axis as printed.
2. The axis is labelled "% wt Silicon", but the compounds sit at 25 (Ti_3Si), 37.5 (Ti_5Si_3), 44.4 (Ti_5Si_4), 50 (TiSi) and 66.7 ($TiSi_2$), their **atomic** percentages of Si; the numbers below are read on the axis as printed, as the key does, so the fractions x are **atom (mole) fractions** and the Phase 4 balances count Si atoms. Ti_5Si_3 has a composition range on the figure: about 35 on its Ti side (1330 and 1170 °C) and up to about 38 on its Si side (1920 °C).
3. Melting is congruent at a maximum of the liquidus and incongruent at a peritectic line [Session 8 Slide 17].

Data read on the diagram

Line	Phases (% Si)
1330 °C (Ti side)	βTi ≈ 5.5, L ≈ 14, Ti_5Si_3 ≈ 35
1170 °C	βTi ≈ 4, Ti_3Si 25, Ti_5Si_3 ≈ 35

PHASE 2 · FRAMES & STRATEGY

Frame: the Ti-Si diagram (% Si, °C); all the work is on horizontal lines and the vertical line of the alloy, so no change of basis is needed. Reactions are named by their shape [Session 8 Slide 6], amounts by the lever rule [Session 7 Slides 16, 32].

PHASE 3 · STEP-BY-STEP DERIVATION

a) Congruent and incongruent melting

- **Congruent:** Ti_5Si_3 (top of the large dome, 2130 °C) and $TiSi_2$ (small maximum near 1500 °C).
- **Incongruent:** Ti_5Si_4 and TiSi end at peritectic lines (1920 and 1570 °C), and Ti_3Si at the peritectoid line (1170 °C): on heating each decomposes into another phase plus liquid (or, for Ti_3Si, into two solids).

b) Invariant reactions (on cooling, % Si read on the axis)

T (°C)	Reaction	Type
1920	$L\,(\approx 48) + \mathrm{Ti_5Si_3}\,(\approx 38) \rightarrow \mathrm{Ti_5Si_4}\,(\approx 44)$	peritectic
1570	$L\,(\approx 59) + \mathrm{Ti_5Si_4}\,(\approx 44) \rightarrow \mathrm{TiSi}\,(50)$	peritectic
1480	$L\,(\approx 64) \rightarrow \mathrm{TiSi}\,(50) + \mathrm{TiSi_2}\,(\approx 67)$	eutectic
1330	$L\,(\approx 14) \rightarrow \beta\mathrm{Ti}\,(\approx 5.5) + \mathrm{Ti_5Si_3}\,(\approx 35)$	eutectic
1330	$L\,(\approx 84) \rightarrow \mathrm{TiSi_2}\,(\approx 67) + (\mathrm{Si})\,(\approx 100)$	eutectic
1170	$\beta\mathrm{Ti}\,(\approx 4) + \mathrm{Ti_5Si_3}\,(\approx 35) \rightarrow \mathrm{Ti_3Si}\,(25)$	peritectoid
865	$\beta\mathrm{Ti}\,(\approx 1) \rightarrow \alpha\mathrm{Ti}\,(\approx 0.5) + \mathrm{Ti_3Si}\,(25)$	eutectoid

c) Cooling of Ti-30 % Si to 600 °C

1. Liquid down to the liquidus of Ti5Si3 (roughly 1950 °C, a reading); there **primary Ti5Si3** starts to crystallise.
2. L + Ti5Si3 down to 1330 °C; the liquid becomes poorer in Si along the liquidus and reaches 14 % Si.
3. At 1330 °C the remaining liquid gives the **eutectic** βTi + Ti5Si3.
4. Between 1330 and 1170 °C: primary Ti5Si3 in a eutectic of βTi + Ti5Si3.
5. At 1170 °C the **peritectoid** βTi + Ti5Si3 → Ti3Si uses up all the βTi (the alloy, 30 %, lies between Ti3Si, 25 %, and Ti5Si3, 35 %), so below it the alloy is **Ti3Si + Ti5Si3**.
6. No other line is crossed down to 600 °C; the boundaries of both compounds are vertical, so nothing changes.

At 1329 °C (βTi 5.5 / Ti5Si3 35, length 29.5):

$$x_{\beta\mathrm{Ti}} = \frac{35 - 30}{35 - 5.5} = \frac{5}{29.5} = 0.17 \qquad x_{\mathrm{Ti_5Si_3}} = \frac{30 - 5.5}{29.5} = \frac{24.5}{29.5} = 0.83$$

At 1169 °C and at 600 °C (Ti3Si 25 / Ti5Si3 35, length 10):

$$x_{\mathrm{Ti_3Si}} = \frac{35 - 30}{35 - 25} = \frac{5}{10} = 0.5 \qquad x_{\mathrm{Ti_5Si_3}} = \frac{30 - 25}{10} = 0.5$$

d) Microconstituents At 1329 °C: primary Ti5Si3 and eutectic. Short tie line from the eutectic liquid (14) to Ti5Si3 (35), length 21:

$$x_{\mathrm{Ti_5Si_3},p} = \frac{30-14}{35-14} = \frac{16}{21} = 0.76 \qquad x_E = \frac{35-30}{21} = \frac{5}{21} = 0.24$$

At 1169 °C: the peritectoid has turned the βTi (which was inside the eutectic) and part of the Ti5Si3 into Ti3Si; the structure is Ti3Si and Ti5Si3, in the same amounts as the phases, 0.5 and 0.5.

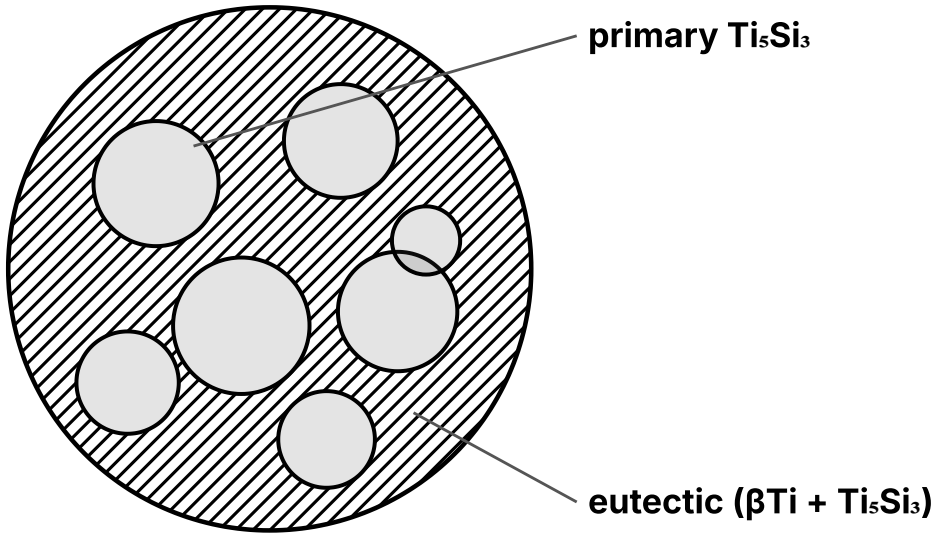


Figure 21. Ti-30 wt% Si just below 1330 °C: primary Ti_5Si_3 (76 %) in the eutectic βTi + Ti_5Si_3 (24 %).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- 1329 °C: $0.17 \times 5.5 + 0.83 \times 35 = 0.94 + 29.06 = 30.0$ at.% Si (atom balance).
- 1169 °C: $0.5 \times 25 + 0.5 \times 35 = 30$ at.% Si.
- Eutectic at 1329 °C: it holds all the βTi (0.17) and Ti5Si3 $0.83 - 0.76 = 0.07$; $0.17 + 0.07 = 0.24 = x_E$.
- **Limiting cases** (atom fractions, dimensionless; compositions in at.% Si as printed): at 1329 °C, as $C_0 \to 14$ the primary Ti5Si3 tends to 0 and as $C_0 \to 35$ to 1; 30 lies close to the compound end, hence 0.76 of primary Ti5Si3.

Interpretation The Ti-rich side combines a eutectic, a peritectoid and a eutectoid. The 30 % alloy ends up made only of two silicides, which are hard and brittle compounds.

Check against the official solution

All values agree with the key (0.17/0.83, 0.5/0.5, 0.24/0.76; the classification of the compounds and the seven reactions).